

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
 - (b) Design a responsive navigation bar using appropriate CSS techniques.
 - (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
 - (d) Evaluate the usability and suggest improvements for better user experience.
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Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
 - (b) Suggest appropriate **CSS animation or transition** for product interaction.
 - (c) Design a hover effect for a product card.
 - (d) Evaluate performance and user experience considerations.
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Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).
- (b) Compare **Flexbox and Grid** based on layout needs.
- (c) Choose the most suitable layout technique and justify your choice.

(d) Evaluate the scalability and responsiveness of the chosen approach.

Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
 - (b) Design a mobile-first layout using CSS.
 - (c) Demonstrate how the layout scales for larger screens using media queries.
 - (d) Evaluate accessibility and readability of the design.
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Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

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CO-2

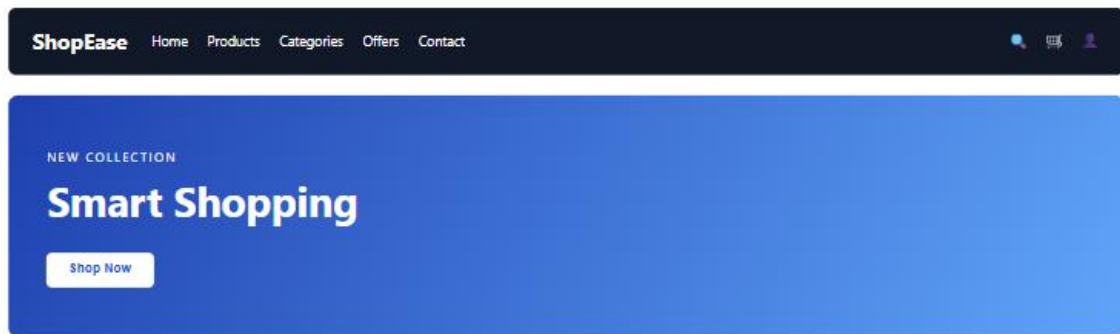
QUESTION 1

Responsive Navigation Bar Design

4 Marks

E-commerce website requiring a responsive navigation bar that adapts across desktop, tablet and mobile devices.

Answer: E-Commerce Website



Resize the browser window to see the navigation automatically adapt to desktop, tablet and mobile.

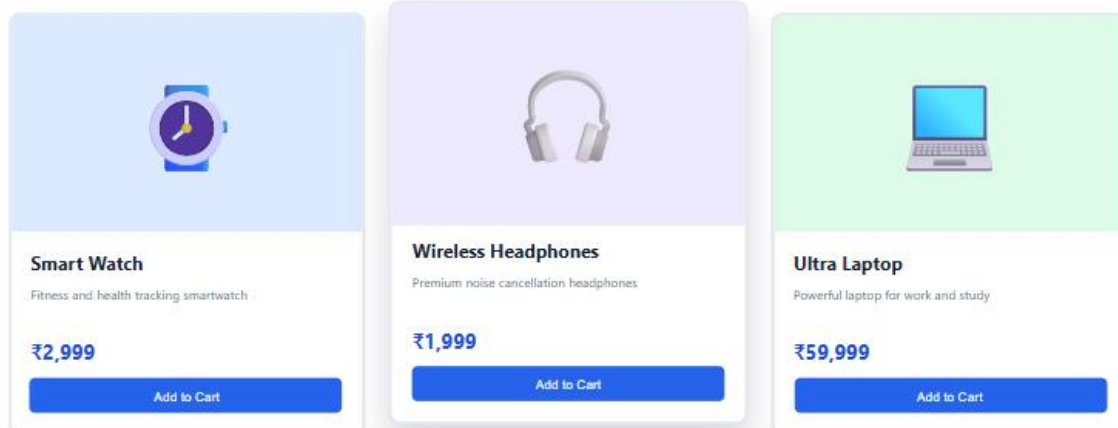
QUESTION 2

Product Hover Effect

4 Marks

E-commerce website requiring visual effects on product items to improve user interaction.

Answer: Product Gallery



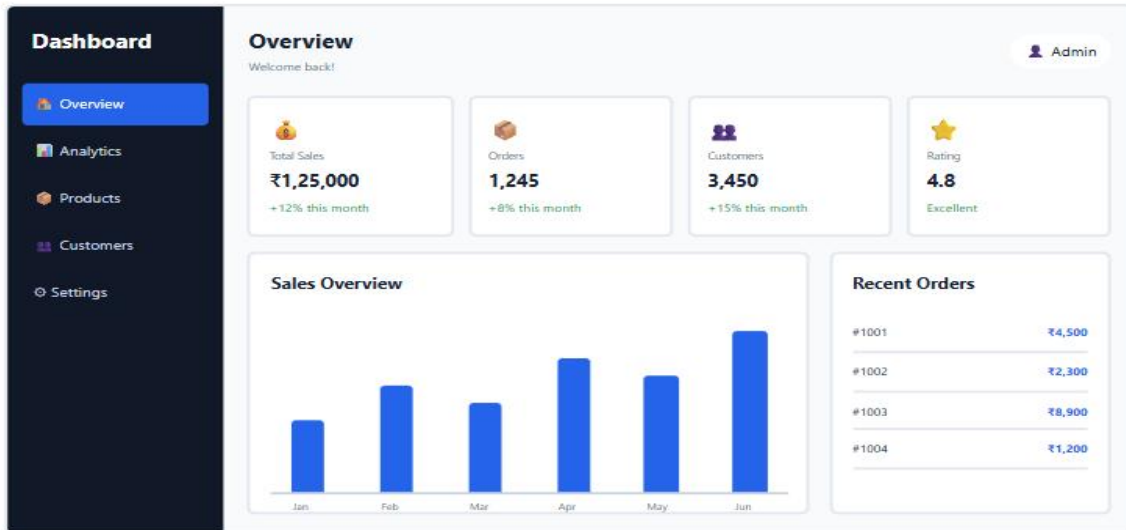
Move your mouse over the product cards to see the hover effect.

QUESTION 3 Layout Selection – Flexbox vs Grid

4 Marks

Dashboard layout requiring rows, columns and proper alignment.

Answer: Responsive Dashboard using CSS Grid

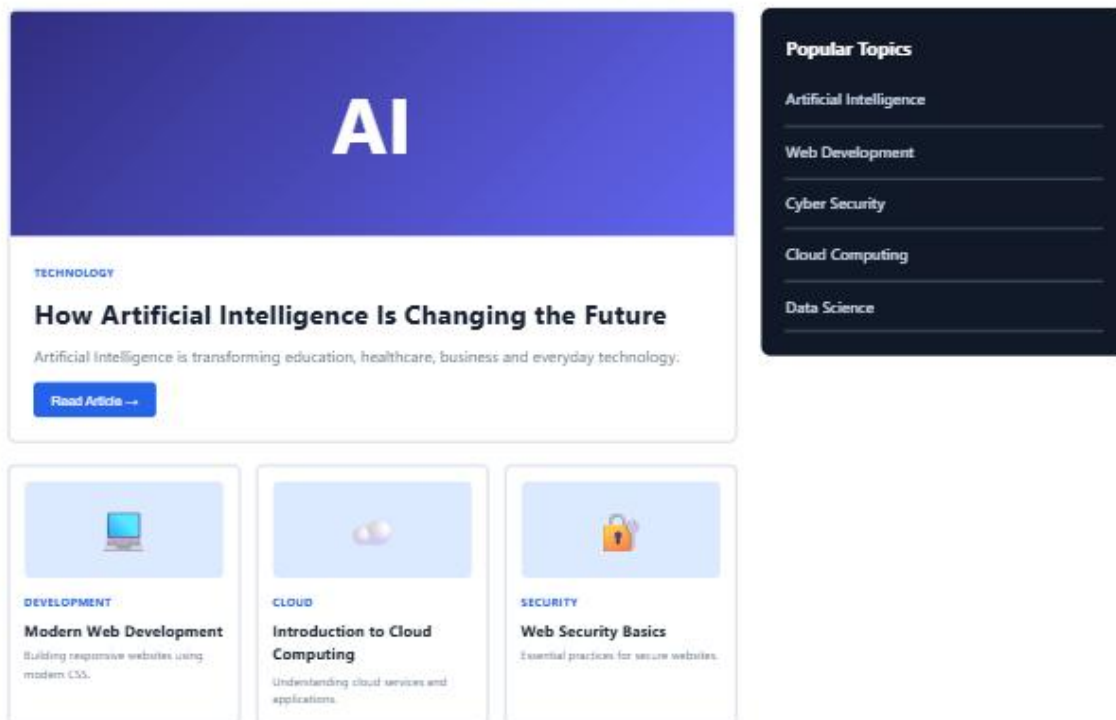


QUESTION 4 Mobile-First Blog Layout

4 Marks

Blog page designed following a mobile-first approach.

Answer: Responsive Tech Blog



This layout starts as a single-column mobile layout and expands into multiple columns on larger screens.

QUESTION 5

JavaScript Event Handling for Dynamic Menu

4 Marks

Website requiring a dynamic menu that responds to user interaction.

Answer: Interactive JavaScript Menu

